

COLEGIO DE MONTALBAN

Integrated Library Management System (ILMS) · Student Mobile Companion

Document Type: Official Mobile Application Feature Specification & User Guide

Target Audience: Capstone Defense Panel, Head Librarian, CDM Students & Faculty

Live Simulator URL: <http://localhost:5175/> | **Desktop Portal:** <http://localhost:5173/>

1. Executive Summary & Application Scope

The **CDM Student Library Mobile App** serves as the dedicated digital companion for enrolled students of Colegio de Montalban. Built with high-performance responsive web and native-mobile compatibility, it empowers students to browse institute-curated references, track loan due dates, reserve books remotely, monitor clearance holds, and access a virtual biometric/QR library pass for express kiosk check-in.

The mobile platform operates in seamless real-time synchronization with the central SQLite database and Express REST API backend, providing automated push alerts, penalty calculations, and transparent lost book physical replacement tracking.

2. Updated Mobile Features & Capabilities Matrix

Module / Feature	Core Functionality	Benefit to Students & Library Staff
1. Digital Library Pass & QR Barcode	Dynamic virtual student ID with live QR code for instant scanner check-in at library turnstiles and circulation counter.	Eliminates paper borrower cards and prevents impersonation; verifies active enrollment in real-time.
2. Multi-Institute Catalog & Curriculum Filter	Curated browsing categorized by Institute (ICS, IBE, IED, ICJ, GENED), Year Level (1st-4th), Semester (1st-2nd), and Dewey/LC Call Numbers.	Allows students to quickly locate exact syllabus-prescribed books and physical shelf locations.
3. MARC 21 Bibliographic Data Viewer	Standardized MARC 21 tagged display (020, 082, 100, 245, 260, 650, 990) with copy-to-clipboard bibliographic support.	Provides official academic citations and metadata compliant with CHED and library accreditation standards.
4. Lost Book Physical Replacement Tracker	In-app amber alert displaying exact Title, Author, and Required ISBN for mandatory same-copy physical replacement. Zero cash markup on book cost.	Guarantees transparent policy compliance and preserves library physical shelf collection integrity.
5. Active Loans & Penalty Countdown	Real-time loan status countdown (Due in 2 days, Overdue) with automated ₱5.00/day late penalty computation.	Prevents overdue accrual and ensures students know exact return dates and financial liabilities.
6. Remote Reservation & Pickup Date Picker	2-step instant book reservation with student-selected target pickup dates and queue status notifications.	Saves students physical travel time and guarantees book reservation holding at the circulation desk.
7. Smart Notification & Clearance Hub	Automated notification feed for due dates, pickup readiness, return receipts, and semester clearance account holds.	Eliminates graduation and enrollment clearance delays by proactively notifying students of account holds.
8. Thematic Visuals & Dark Mode UX	Subject-matched cover graphics (Ethics, Code, Forensics, Business, Pedagogy) and one-tap high-contrast Dark / Light theme toggle.	Delivers an engaging, premium user experience aligned with official Colegio de Montalban branding.

3. Deep-Dive Feature Specifications

A. Digital Student Library Pass & Access Control

- **Biometric / QR Code Generation:** Generates unique, secure student tokens mapped to student institutional IDs (e.g., 2024-0042).
- **Account Clearance Status:** Real-time badge shows 'Active & Cleared' or 'Account Hold (Overdue / Pending Replacement)'.
- **Kiosk Compatibility:** Scan-ready under optical barcode/QR scanners at the CDM library entrance.

B. Curated Catalog & MARC 21 Bibliographic Standards

- **Curriculum-Aligned Collections:** 160+ curated textbook references categorized by Institute (ICS, IBE, IED, ICJ, GENED) with 5 books per semester per year level.
- **Dewey Decimal / LC Call Numbers:** Displays shelf location codes (e.g., 004.01 P39c 2002, 657.046 V23f 2021).
- **Full MARC 21 Tagging:** Tag breakdown includes MARC 020 (ISBN), MARC 082 (Classification), MARC 100 (Author), MARC 245 (Title), MARC 260 (Imprint), MARC 650 (Subject), and MARC 990 (Curriculum).

C. Lost Book Physical Replacement Policy Workflow

Official CDM Physical Replacement Rule: In compliance with library audit guidelines, lost materials cannot be settled via cash replacement fees for the book value. Borrowers must surrender an identical physical copy (matching Title, Author, Edition, and ISBN) plus settle any accrued daily late fines (₱5/day). The mobile app provides a dedicated tracking card to guide the student.

D. Remote Reservation, Loan Management & Notifications

- **Remote Hold Request:** Students select a book and specify their target pickup date.
- **Shelf Availability Countdown:** Real-time inventory tracking (e.g., '3 / 5 Available') prevents wasted trips to the library.
- **Automated Loan Alerts:** Push reminders triggered 48 hours and 24 hours prior to due dates, plus instant notifications upon desk check-in.

4. Technical Stack & Deployment Endpoints

Component	Technology / Platform	Live Port / URL
Student Mobile App	React, Vite, Lucide Icons, Vanilla CSS	http://localhost:5175/
Librarian Desktop Portal	React, TypeScript, TailwindCSS, Figma UI	http://localhost:5173/
Backend REST API	Node.js, Express, Better-SQLite3, JWT, Bcrypt	http://localhost:5002/api/books
Public Promo & OPAC	HTML5, CSS3, Vanilla JS Web Portal	http://localhost:5002/promo